



The Solar Cook Book

Professional Solar Guide for Jamaica

From phone-charging kits to
10kW+ home arrays

Learn to size, wire, and build
solar systems

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Welcome to Solar DIY

Solar energy is no longer just for large corporations or wealthy homeowners. With falling panel prices and improved battery technology, anyone can build their own solar power system.

Getting Started

This guide covers systems from 100W portable kits to 10kW+ home arrays. All examples are tailored for Jamaica's climate with 5.0-5.5 peak sun hours daily.



What You'll Learn

Panels

Monocrystalline
vs poly,
wattage sizing,
wiring configs

Batteries

LiFePO4 vs
lead-acid,
capacity math,
runtime

Controllers

PWM vs MPPT,
voltage
matching,
efficiency

Inverters

Pure sine wave
sizing, surge
loads, AC
distribution

Small Systems

(100W-500W)

Ideal for phone charging, laptop charging, LED TVs, and small electronics.

System	Use Case	Cost Estimate
100W Portable Kit	Charge 2 phones + 1 laptop daily; run LED TV 2 hours	50-50 USD
200W Starter	Adds headroom for cloudy days. Great for sheds and RVs	00-50 USD
400W Cabin Kit	Small off-grid setup with inverter for	00-00 USD

	lights, fans, charging	
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Pro Tip: Start small. A 100W kit teaches you the basics without a big investment. You can always expand later.

Medium Systems

(500W-2kW)

Perfect for Starlink Mini, larger TVs, multiple laptops, and home office setups.

System	Use Case
800W Office	Power 2 laptops + TV + router during outages. Great for remote work.
1.2kW Hybrid	Adds battery backup for evening use. Supports JPS net-billing if grid-tied.
1.8kW Expandable	Room to add more panels later. Can handle small appliances plus devices.

Key Components

Panels:

400W-1kW
monocrystalline
array

Battery: 12V

100Ah-200Ah
LiFePO4 bank

Controller:

40A MPPT
charge
controller

Inverter:

1000W-
2000W pure
sine wave

Large Systems (2kW-10kW+)

Full home power: multiple TVs, laptops, Starlink Mini, fridge, and all devices.

System	Use Case
3kW Home Backup	Covers essential circuits: lights, wifi, fridge, TV, and laptop loads.
5kW Whole-Home	Run most household circuits. Ideal for frequent outages or full off-grid.
10kW+ Estate	Multiple inverters, large battery bank, commercial-grade installation.

Professional Installation by Nash Technologies: Large systems require certified electricians. Nash Technologies

provides full installation, JPS net-billing support, 15-year warranties, free site assessments, and custom system design for homes and businesses across Jamaica.



Safety First

Electrical Safety Warning: Solar systems produce high voltage DC that can cause serious injury or fire. Follow NEC code, use properly sized wire, install fuses/breakers on all conductors, and hire a licensed electrician for grid-tied connections.

Wire Sizing & Fusing

Undersized wiring overheats. Size conductors for **1.25× expected current**. Fuse every positive conductor at the battery and between components. Use marine-grade or solar-rated cable for outdoor runs.

NEC Requirements

Disconnects:

DC and AC disconnects required

Grounding:

All frames and enclosures grounded

Labeling:

Warning signs at all junction boxes

Inspections:

JPS inspection for grid-tied systems

Custom Builds & Blueprints

Step-by-step blueprints for small units. Use these as a starting point and adjust to your needs.

Build 1: 100W Phone Charging Kit

Skill Level: Beginner | **Time:** 30 minutes | **Cost:** ~7,700 JMD

Parts List

Part	Spec	Qty	Est. Price (JMD)
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Solar Panel	20W monocrystalline, 12V	1	,000
Battery	12V 7Ah LiFePO4	1	,500
Charge Controller	10A PWM	1	,500
USB Output Board	5V USB-A dual port	1	,200
Wiring Kit	16AWG solar cable, MC4 connectors	1	,500
Enclosure	Waterproof project box	1	00
Total			~8,000 JMD

Wiring Diagram

**Panel (+) → Controller (+) →
Battery (+) → Load (+) →
Ground**

Panel (-) → Controller (-) → Battery (-)
→ Load (-)

Step-by-Step Build

1. Mount Panel

Place in full sun, angled toward equator

2. Wire Panel to Controller

Red to +, black to -, keep runs short

3. Connect Battery

Connect battery FIRST before panel

4. Wire USB Output

5V board to battery terminals

5. Test & Seal

Verify charging, mount in enclosure

6. Deploy

Place in sun, charge devices

Build 2: 200W

Laptop Station

Skill Level: Intermediate | **Time:** 1-2 hours | **Cost:** ~3,000 JMD

Parts List

Part	Spec	Qty	Est. Price (JMD)
Solar Panel	100W monocrystalline, 12V	2	4,000
Battery	12V 50Ah LiFePO4	1	8,000
Charge Controller	20A MPPT	1	,000
Inverter	300W pure sine wave	1	2,000
Wiring + Breakers	12AWG solar cable, 20A fuse	1	,000
		1	,000

Enclosure / Rack	Steel or aluminum frame		
Total			~0,000 JMD

Wiring Diagram

2× Panels in Parallel → 20A MPPT → 12V 50Ah LiFePO4 → 300W Inverter → AC Outlets

Fuse on positive battery cable. Ground all metal frames.

Build 3: 400W Cabin Kit

Skill Level: Intermediate | **Time:** 2–3 hours | **Cost:** ~5,000 JMD

Parts List

Part	Spec	Qty	
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			Est. Price (JMD)
Solar Panel	200W monocrystalline, 12V	2	6,000
Battery	12V 100Ah LiFePO4	1	5,000
Charge Controller	40A MPPT	1	4,000
Inverter	1000W pure sine wave	1	8,000
Wiring + Breakers	10AWG solar cable, 40A fuse	1	,000
Mounting Rack	Roof or ground mount	1	,000
Total			~26,000 JMD

Wiring Diagram

**2 × 200W Parallel → 40A MPPT
 → 12V 100Ah → 1000W Inverter
 → AC Distribution Panel**

Use 40A breaker between controller and battery. Install DC disconnect.

DIY Device Charging Systems

Step-by-step build guides for charging phones, laptops, Starlink Mini, and TVs with solar.

Phone & Tablet Charging Kit

- **Panel:**
20W
monocrystalline

- **Battery:**
12V 7Ah
LiFePO4
- **Controller:**
10A PWM
- **Output:**
5V USB-A
ports
- **Runtime:**
2–3 phones
per day

Laptop Charging Station

- **Panel:**
100W
monocrystalline
- **Battery:**
12V 50Ah
LiFePO4
(~640Wh)
- **Controller:**
20A MPPT

- **Inverter:**

300W pure
sine wave

- **Runtime:**

2-4 hours
laptop use

More Device Setups

Starlink Mini Power

- **Panel:**
120W
monocrystalline
- **Battery:**
12V 100Ah
LiFePO4
- **Controller:**
20A-30A
MPPT
- **Inverter:**
300W pure
sine wave

- **Runtime:**
6–10 hours
off-grid

TV & Entertainment

- **Panel:**
150W
monocrystalline
- **Battery:**
12V 100Ah
LiFePO4
- **Controller:**
20A MPPT
- **Inverter:**
500W pure
sine wave
- **Runtime:**
3–6 hours
for 55" LED

Solar & Electrical Glossary

Simple definitions for common terms on solar labels, invoices, and JPS net-billing paperwork.

Term	Definition
Watt (W)	Unit of power. A 100W bulb uses 100 watts while on.
Kilowatt (kW)	1,000 watts. A 5 kW system produces 5,000 watts

	at peak sun.
Kilowatt-hour (kWh)	Unit of energy. JPS bills in kWh.
Volt (V)	Electrical pressure. Common solar banks: 12V, 24V, 48V.
Ampere (A)	Current flow. Wiring and fuses are sized by amps.
Amp-hour (Ah)	Battery capacity. A 100Ah battery delivers 5A for 20 hours.
PSI	Pounds per square inch. Used for water

	pressure and pipe fittings.
BTU	British Thermal Unit. Heat energy; useful for water-heating sizing.
Peak Sun Hours	Hours of 1,000 W/m ² irradiance. Jamaica averages 5.0–5.5.
Inverter	Converts DC battery power to AC for appliances.
Charge Controller	Regulates panel-to-battery charging to prevent overcharge.

Grid-Tied	Connected to JPS lines. Allows net billing for excess energy.
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Solar Calculator

Estimate daily energy production and system cost for Jamaica conditions.

Formula:

$$\text{Panel Watts} \times \text{Count} \times \text{Sun Hours} \times \text{Efficiency} \div 100$$

Input	Value	
Panel Watts	400W	
Panel Count	4	
Sun Hours (Jamaica)	5.5	

Efficiency	75%
Daily Output	8,800 Wh
Estimated Cost	,280 USD

Use this formula to size any system.
Adjust panel count and wattage to match your energy needs.

Solar Parts Finder

Select a system to see the full bill of materials with local-market pricing in JMD.

Component	Phone/ Tablet	Laptop Station	Starlink Mini
Panel	20W — ,000 JMD	100W — 2,000 JMD	120W — 4,000 JMD
Battery	12V 7Ah LiFePO4 — ,500 JMD	12V 50Ah LiFePO4 — 8,000 JMD	12V 100Ah LiFePO4 — 5,000 JMD
Controller	10A PWM —	20A MPPT —	20A-30A MPPT —

	,500 JMD	,000 JMD	4,000 JMD
Inverter	—	300W — 2,000 JMD	300W — 2,000 JMD
Other	5V USB-A — ,200 JMD Wiring — ,500 JMD	Wiring + breakers — ,000 JMD	DC-DC converter — ,000 JMD
Total	~7,700 JMD	~3,000 JMD	~0,000+ JMD

For full home
systems and custom
builds, contact
Microtuff Solutions
for a detailed quote.

Nash Technologies

Professional
Solar
Installation
Partner

**Trusted
Installer:**
Nash
Technologies
is a certified
solar
installation
company
serving all 14
parishes of
Jamaica.
Specializing in

residential,
commercial,
and industrial
solar power
systems.

Services

Residential Solar

Complete home solar systems from 3kW to 10kW+

Commercial Installation

Office buildings, warehouses, and retail spaces

JPS Net-Billing

Full paperwork and interconnection support

System Design

Custom layouts, equipment selection, and load analysis

Maintenance **Warranty Support**

Panel
cleaning,
inverter
service,
battery
checks

15-
year
warranties
on
workmanship
and
equipment

Why Choose Nash Technologies?

- **Certified
Installers:**

Trained
and
licensed by
Jamaica
Public
Service
(JPS)

- **Free Site Assessments:**

No-obligation evaluation of your property and energy needs

- **Quality Equipment:**

Only tier-1 panels, LiFePO4 batteries, and MPPT controllers

- **Local Support:**

Based in Mandeville, serving all parishes

- **15-Year Warranties:**

Peace of mind on

every
installation

- **JPS**

- Interconnection:**

- Handle all
permits,
inspections,
and net-
billing
agreements

Contact Nash Technologies

Phone: **Service**

1(876) Area:

359-99

All

14

Parishes,
Jamaica

Headqu **Email:**

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Jamaica

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**Ready to go
solar?**

Call or
WhatsApp
Nash
Technologies
today for a
free
consultation

and site
assessment.

Sourcing Parts in Jamaica

Where to find solar
components locally
and what to look for
when buying.

Local Suppliers

Hardware Stores Electrical Wholesalers

Banks, Kingston
Jamaica and
Producers Montego
Fontana Bay
for have
basic specialized
wiring, electrical
fuses, suppliers
enclosures for
and breakers,
tools. panels,
and
inverters.

Solar Retailers

Mandeville and Kingston solar shops carry panels, batteries, and controllers.

Online Marketplaces

Facebook groups, Jamaican classifieds, and regional distributors.

What to Buy Locally vs Import

Buy Locally:

Wiring, fuses, breakers, enclosures, basic tools, MC4

connectors,
cable trays.

**Consider
Importing:**

High-
efficiency
panels,
premium
LiFePO4
batteries,
MPPT
controllers,
pure sine
wave
inverters.

Quality Checklist

- **Panels:**
Look for
Tier-1
manufacturers,
check
warranty

(25-year performance), verify IEC 61215 certification.

- **Batteries:**

LiFePO₄ preferred over lead-acid. Check cycle life (2000+ cycles), warranty (5-10 years), and BMS included.

- **Controllers:**

MPPT over PWM for systems over 200W. Verify voltage range matches

your
panels.

- **Inverters:**

Pure sine
wave only
for
sensitive
electronics.
Check
surge
rating and
efficiency
($>90\%$).

- **Wiring:**

Use solar-
rated cable
(PV wire),
proper
AWG for
current,
and UV-
resistant
outer
sheath.

Budget Tips

Start Small

Buy a 100W kit first to learn before investing big.

Buy in Bulk

Group buys with neighbors can cut costs 15–20%.

Used Panels

Used or B-grade panels are cheaper and still functional.

Local Assembly

Buy parts separately and assemble yourself to save labor.

Need Help Sourcing?

Contact
Microtuff
Solutions for a
curated parts
list and
vendor
recommendations.
We can also
coordinate
bulk orders for
community
projects.

Need Help?

Professional
installation, system
design, and JPS
net-billing support
across Jamaica.

Nash Microtuff Techn Solutions

Certified solar installation, 15-year warranty, free site assessment. Security research, pentesting, FRP removal, hardware sales.

**1(876)
803-6062**

**1(876)
359-9983**

Ready to start your solar project?

Contact us for a free consultation and site assessment. We serve all

14 parishes
across
Jamaica.

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This guide is for
educational
purposes. Always
consult a licensed
electrician for
installation.